GOOBO LABS · LESSON

Sending Messages Over the Internet

How your emails, photos and messages travel safely across the network.

LEARNING OBJECTIVES

After this lesson, you will understand

- What digital data is
- What packets are
- What a server does
- How packets travel from sender to receiver
- How messages travel over the Internet
- What a router does
- Why packets are numbered

• INTRODUCTION

Every day we send emails, WhatsApp messages, photos, videos, and files.

To a computer, all of these are simply digital data.

The Internet follows a set of rules to send this data safely from one device to another.

DIGITAL DATA

A computer stores everything as digital data.

Text

Images

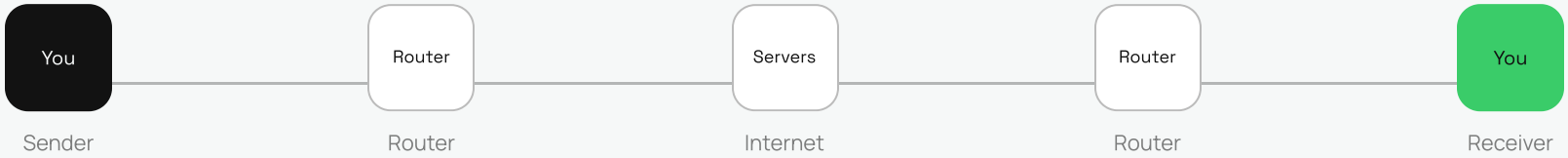
Audio

Videos

All of these are stored using **0s and 1s**

HOW A MESSAGE TRAVELS

The message passes through many devices before it arrives.



WHAT IS A PACKET?

A packet is a small piece of a message.

Instead of sending one large message, the computer divides it into many small packets — automatically.

"Hello Omar, welcome to Goobo Labs."

splits into

1 · Hello

2 · Omar,

3 · welcome

4 · to

5 · Goobo

6 · Labs

WHY SPLIT MESSAGES?

Splitting a message into packets is:

- Easier to send
- Faster
- More reliable
- Better network performance

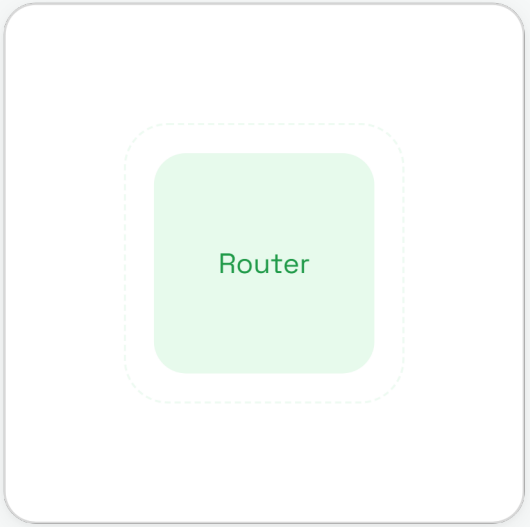
Large files may contain thousands of packets.

THE ROUTER

A router connects your network to the Internet.

Every packet passes through the router before entering the Internet.

- Connect devices
- Forward packets
- Choose the next network





THE SERVER

A server helps move packets across the Internet.

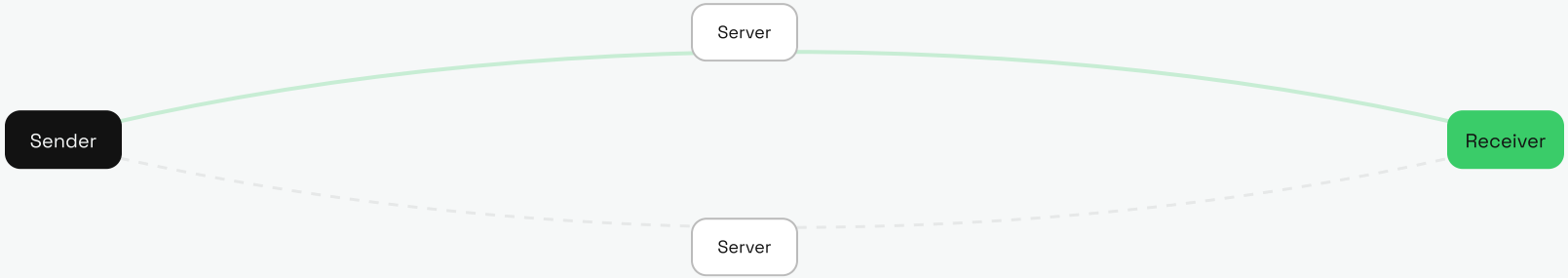
Each server receives a packet, then decides where to send it next.

Packets keep moving from server to server until they reach the destination.

HOW DO PACKETS TRAVEL?

There is no fixed route.

Every server chooses the best available path. If one route is busy, another is used.

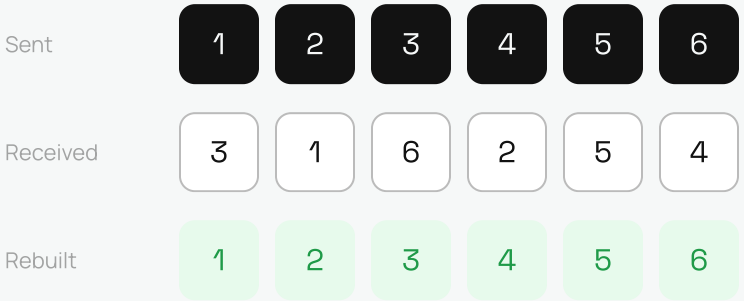


This makes communication faster and more reliable.

WHY ARE PACKETS NUMBERED?

Packets may arrive in a different order.

The receiver uses sequence numbers to rebuild the original message correctly.



REAL-LIFE EXAMPLES

Every one of these uses packets, routers and servers.

- Email
- WhatsApp
- Facebook Messenger
- Video streaming
- File downloads
- Web browsing

KEY TERMS

TERM	MEANING
Digital Data	Information stored as 0s and 1s
Packet	A small piece of a message
Router	Connects a network to the Internet
Server	Forwards packets across the Internet
Sender	Device sending the message
Receiver	Device receiving the message

SUMMARY

- All Internet communication uses digital data.
- Messages are divided into packets.
- Every packet passes through a router.
- Servers forward packets across the network.
- Packets may take different routes.
- Packets are numbered so they can be rebuilt correctly.

PRACTICE QUESTIONS

1. What is digital data?
2. What is a packet?
3. Why are messages divided into packets?
4. What is the job of a router?
5. What is the job of a server?
6. Why are packets numbered?
7. Can packets travel through different routes?

30-SECOND EXPLANATION

When we send a message over the Internet, the computer converts it into digital data and divides it into small packets. Every packet passes through a router and travels across different servers, each choosing the best path to the destination. Because packets may arrive out of order, each one carries a sequence number so the receiver can rebuild the message correctly.